

Formative 3 — Group Contributions

Probability Distributions, Bayesian Probability & Gradient Descent

Team: David Shumbusho · Samuel Niyonkuru · Ishimwe Shumbusho Anglebert · Tifare Elegance Kaseke

Team Members — Role / Responsibility

Team Member	Role / Responsibility	Email
David Shumbusho	Part 1 – EM Algorithm + Manual Gradient Descent Update 1	d.shumbusho@alustudent.com
Samuel Niyonkuru	Part 2 – Bayesian Probability + Manual Gradient Descent Update 2	s.niyonkuru@alustudent.com
Ishimwe Shumbusho Anglebert	Manual Gradient Descent Update 3 + Checking calculations	a.ishimwe6@alustudent.com
Tifare Elegance Kaseke	Part 4 – Gradient Descent in Python + Manual Gradient Descent Update 4 + GitHub / README	t.kaseke@alustudent.com

Task Allocation and Tracker

Task Allocated	Assigned Member	Deadline	Completion Status	Reviewed by Team?	Comments
Load and inspect Galton Families height dataset	David Shumbusho	02 Jul 2026	Completed	Yes	Prepared height data for the EM Algorithm section.
Implement EM Algorithm from scratch	David Shumbusho	03 Jul 2026	Completed	Yes	Worked on Gaussian probability, E-step, M-step, and parameter updates.
Explain why splitting at the global mean is not reliable	David Shumbusho	03 Jul 2026	Completed	Yes	Explained that height groups overlap, so EM is better because it uses soft probabilities.
Produce EM tracking table and test-height posterior output	Ishimwe Shumbusho Anglebert	03 Jul 2026	Completed	Yes	Helped produce the required EM table and posterior probability output for a random test height.
Complete manual gradient descent setup and Update 1	David Shumbusho	04 Jul 2026	Completed	Yes	Wrote the given matrices, initial prediction, error, MSE, gradients, and first update.
Load IMDb Movie Reviews dataset	Samuel Niyonkuru	02 Jul 2026	Completed	Yes	Prepared review and sentiment data for Bayesian probability.
Select positive and negative sentiment keywords	Samuel Niyonkuru	03 Jul 2026	Completed	Yes	Selected keywords such as great, excellent, amazing, love, bad, boring, terrible, and waste.
Implement Bayes' Theorem using basic Python	Samuel Niyonkuru	03 Jul 2026	Completed	Yes	Calculated prior, likelihood, marginal, and posterior probabilities.
Explain Bayesian probability results	Samuel Niyonkuru	03 Jul 2026	Completed	Yes	Explained what $P(\text{Positive} \mid \text{keyword})$ means and how

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					keywords affect sentiment probability.
Complete manual gradient descent Update 2	Samuel Niyonkuru	04 Jul 2026	Completed	Yes	Continued from Update 1 values and calculated prediction, error, MSE, gradients, and updated m and b.
Complete manual gradient descent Update 3	Ishimwe Shumbusho Anglebert	04 Jul 2026	Completed	Yes	Continued from Update 2 values and calculated the third update.
Review manual gradient descent calculations	Ishimwe Shumbusho Anglebert	04 Jul 2026	Completed	Yes	Checked that matrix multiplication, gradients, and update steps were clear and correct.
Complete manual gradient descent Update 4	Tifare Elegance Kaseke	04 Jul 2026	Completed	Yes	Continued from Update 3 values and calculated the fourth update of m and b.
Write final gradient descent trend explanation	Tifare Elegance Kaseke	04 Jul 2026	Completed	Yes	Explained that MSE decreased across updates, proving gradient descent reduced error.
Implement gradient descent in Python	Tifare Elegance Kaseke	05 Jul 2026	Completed	Yes	Converted the manual calculations into Python using matrix operations.
Add SciPy derivative implementation	Tifare Elegance Kaseke	05 Jul 2026	Completed	Yes	Added derivative-related implementation using SciPy as required.
Create Matplotlib plots	Tifare Elegance Kaseke	05 Jul 2026	Completed	Yes	Created plots showing how m, b, and error changed over iterations.
Set up GitHub repository and push files	Tifare Elegance Kaseke	05 Jul 2026	Completed	Yes	Created the repository, pushed notebook, README, datasets, and manual calculation files.
Edit README and organize final submission	Tifare Elegance Kaseke	05 Jul 2026	Completed	Yes	Updated project overview, dataset links, running instructions, and assignment explanation.
Run final notebook and check outputs	Tifare Elegance Kaseke + All Members	05 Jul 2026	Completed	Yes	Confirmed the notebook runs correctly and produces the required outputs.

Meeting Attendance Log

Meeting Date	Agenda	Facilitator	Attendees	Key Discussion Points	Next Steps
01 Jul 2026	Kick-off & task split	Tifare Elegance Kaseke	David, Samuel, Anglebert, Tifare	Reviewed the assignment brief and agreed a four-way split; each member takes one manual gradient-descent update. Confirmed conventions: $\alpha = 0.001$ and a per-sample bias vector.	Create the GitHub repository; download and share both datasets.
02 Jul 2026	Data loading & setup	David Shumbusho	David, Samuel, Anglebert,	Galton and IMDb datasets loaded and inspected. Chose	Implement the EM

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			Tifare	father + child heights for the EM mixture and shortlisted positive/negative sentiment keywords.	algorithm (Part 1) and Bayes' theorem (Part 2).
03 Jul 2026	Part 1 & Part 2 review	Samuel Niyonkuru	David, Samuel, Anglebert, Tifare	EM converges once seeded with k-means (avoids the degenerate variance-collapse spike). Bayesian posteriors clearly separate positive from negative keywords. Tracking table and test-height posterior produced.	Finalise plots; begin the four manual gradient-descent updates.
04 Jul 2026	Manual gradient descent	Ishimwe Shumbusho Anglebert	David, Samuel, Anglebert, Tifare	Completed all four manual updates (one per member). MSE decreased $61 \rightarrow 47.031 \rightarrow 36.382 \rightarrow 28.265$, confirming the method. Calculations cross-checked as a group.	Convert the manual work to Python (Part 4); compile the handwritten pages into one PDF.
05 Jul 2026	Part 4 code, integration & dry run	Tifare Elegance Kaseke	David, Samuel, Anglebert, Tifare	Python gradient descent (matrix ops + SciPy + Matplotlib) reproduces the manual results. Notebook runs end-to-end; README and repository finalised; rehearsed the live classification demo.	Submit the assignment and prepare the presentation.

Note: Deadlines reflect the team's working timeline (02–05 July 2026); adjust to your actual dates before submitting. Each member also provides proof of contribution through their GitHub commits, their handwritten manual-calculation page, and their notebook sections.